

PROBABILITY MASTER GUIDE

Comprehensive Theory, Solved Examples & Practice Questions

1. Theory of Probability

What is Probability?

Probability is the branch of mathematics concerning numerical descriptions of how likely an event is to occur, or how likely it is that a proposition is true. The probability of an event is a number between 0 and 1, where, roughly speaking, 0 indicates impossibility of the event and 1 indicates certainty.

Important Terms

- **Experiment:** An action or operation that produces a distinct result (e.g., tossing a coin).
- **Sample Space (S):** The set of all possible outcomes of an experiment.
- **Event (E):** Any subset of a sample space.
- **Favorable Outcomes:** The outcomes which entail the happening of an event.

The Fundamental Formula

If all outcomes of an experiment are equally likely, the probability of an event E happening is given by:

$$P(E) = (\text{Number of favorable outcomes}) / (\text{Total number of possible outcomes})$$

Key Properties

- For any event E , $0 \leq P(E) \leq 1$.
- The sum of the probabilities of all the elementary events of an experiment is 1.

- The probability of a sure event is 1, and an impossible event is 0.
- Probability of NOT E : $P(\text{not } E) = 1 - P(E)$, often written as $P(E')$ or $P(\bar{E})$.

Standard Scenarios

- 1. Coins:** A single coin has 2 outcomes: Head (H), Tail (T). Tossing n coins gives 2^n possible outcomes.
- 2. Dice:** A standard die has 6 faces (1, 2, 3, 4, 5, 6). Rolling n dice gives 6^n possible outcomes.
- 3. Playing Cards:** A standard deck has 52 cards divided into 4 suits of 13 cards each: Spades (Black), Clubs (Black), Hearts (Red), and Diamonds (Red). Each suit has an Ace, King, Queen, Jack, and numbers 2-10. Face cards are King, Queen, and Jack (total 12 face cards).

2. Solved Examples

Here are 12 fully solved examples covering various concepts of probability.

Example 1: Single Coin Toss

What is the probability of getting a Head when a fair coin is tossed?

Solution:

Sample Space, $S = \{H, T\}$. Total outcomes = 2.

Favorable outcome (Head) = $\{H\}$. Number of favorable outcomes = 1.

$$P(\text{Head}) = 1 / 2$$

Example 2: Two Coins Tossed Simultaneously

Two coins are tossed. Find the probability of getting at least one tail.

Solution:

Sample Space, $S = \{HH, HT, TH, TT\}$. Total outcomes = 4.

Favorable outcomes (at least one tail) = $\{HT, TH, TT\}$. Total favorable = 3.

$$P(\text{At least one tail}) = 3 / 4$$

Example 3: Rolling a Single Die

A die is thrown once. What is the probability of getting a prime number?

Solution:

Sample Space, $S = \{1, 2, 3, 4, 5, 6\}$. Total outcomes = 6.

Prime numbers in the set = $\{2, 3, 5\}$. Total favorable = 3.

$$P(\text{Prime Number}) = 3 / 6 = 1 / 2$$

Example 4: Rolling Two Dice

Two dice are thrown at the same time. Find the probability that the sum of the two numbers appearing on the top is 8.

Solution:

Total possible outcomes = $6 \times 6 = 36$.

Favorable outcomes for sum 8 = $\{(2,6), (3,5), (4,4), (5,3), (6,2)\}$. Total favorable = 5.

$$P(\text{Sum is 8}) = 5 / 36$$

Example 5: Standard Deck of Cards - Face Card

One card is drawn from a well-shuffled deck of 52 cards. Calculate the probability that the card will be a face card.

Solution:

Total number of possible outcomes = 52.

Face cards are Kings, Queens, and Jacks. There are 4 of each, so $3 \times 4 = 12$ face cards.

$$P(\text{Face card}) = 12 / 52 = 3 / 13$$

Example 6: Standard Deck of Cards - Specific Color

A card is drawn from a standard deck. What is the probability of drawing a red Ace?

Solution:

Total possible outcomes = 52.

There are two red Aces in a deck: Ace of Hearts and Ace of Diamonds. Favorable = 2.

$$P(\text{Red Ace}) = 2 / 52 = 1 / 26$$

Example 7: Marbles in a Bag

A bag contains 3 red, 5 black, and 4 white marbles. If a marble is drawn at random, what is the probability that it is not black?

Solution:

Total marbles = $3 + 5 + 4 = 12$.

Number of black marbles = 5. So, probability of drawing a black marble = $5/12$.

Probability of NOT black = $1 - (5 / 12) = 7 / 12$.

(Alternatively, non-black marbles = 3 red + 4 white = 7).

Example 8: Leap Year Problem

Find the probability that a leap year selected at random will contain 53 Sundays.

Solution:

A leap year has 366 days. $366 \text{ days} = 52 \text{ complete weeks} + 2 \text{ extra days}$.

52 weeks will definitely have 52 Sundays. The remaining 2 days can be:

(Mon,Tue), (Tue,Wed), (Wed,Thu), (Thu,Fri), (Fri,Sat), (Sat,Sun), (Sun,Mon).

Total combinations = 7. Favorable (having Sunday) = 2 {(Sat,Sun), (Sun,Mon)}.

$$P(53 \text{ Sundays in Leap Year}) = 2 / 7$$

Example 9: Defective Items

A lot of 20 bulbs contain 4 defective ones. One bulb is drawn at random from the lot. What is the probability that this bulb is defective?

Solution:

Total number of bulbs = 20.

Number of defective bulbs = 4.

$$P(\text{Defective bulb}) = 4 / 20 = 1 / 5$$

Example 10: Letters of a Word

A letter is chosen at random from the word "MATHEMATICS". Find the probability that the letter is a vowel.

Solution:

Total number of letters in "MATHEMATICS" = 11.

Vowels in the word are A, E, A, I. Number of vowels = 4.

$$P(\text{Vowel}) = 4 / 11$$

Example 11: Two Friends' Birthdays

Savita and Hamida are friends. What is the probability that both will have different birthdays? (Ignoring a leap year).

Solution:

Total number of days in a non-leap year = 365.

If Savita's birthday is on any day, Hamida's birthday can be on any of the remaining 364 days for them to be different.

$$P(\text{Different birthdays}) = 364 / 365$$

Example 12: Complementary Events

If the probability of winning a game is 0.3, what is the probability of losing it?

Solution:

Let E be the event of winning. $P(E) = 0.3$.

Probability of losing is the complement of winning, $P(\text{not } E)$.

$$P(\text{not } E) = 1 - P(E) = 1 - 0.3 = 0.7$$

3. Practice Questions

Test your knowledge with these 25 multiple-choice questions. Select the correct option for each. The answer key is provided at the end of the document.

Q1. The probability of a sure event is:

- | | |
|--------|-------|
| A) 0 | B) 1 |
| C) 0.5 | D) -1 |

Q2. The probability of an impossible event is:

- | | |
|----------------|-------------|
| A) 1 | B) 0 |
| C) Not defined | D) Infinity |

Q3. Which of the following cannot be the probability of an event?

- | | |
|------------------|---------|
| A) $\frac{2}{3}$ | B) -1.5 |
| C) 15% | D) 0.7 |

Q4. If $P(E) = 0.05$, then what is the probability of 'not E'?

- | | |
|---------|---------|
| A) 0.95 | B) 0.05 |
| C) 1.05 | D) 0.9 |

Q5. A coin is tossed once. What is the probability of getting a tail?

- | | |
|------|------------------|
| A) 1 | B) $\frac{1}{2}$ |
| C) 2 | D) 0 |

Q6. Two coins are tossed simultaneously. What is the probability of getting exactly one head?

- | | |
|------------------|------------------|
| A) $\frac{1}{4}$ | B) $\frac{1}{2}$ |
| C) $\frac{3}{4}$ | D) 1 |

Q7. A die is thrown once. What is the probability of getting a number greater than 4?

- | | |
|------------------|------------------|
| A) $\frac{1}{2}$ | B) $\frac{1}{3}$ |
| C) $\frac{2}{3}$ | D) $\frac{1}{6}$ |

Q8. A die is thrown once. The probability of getting an even number is:

- | | |
|------------------|------------------|
| A) $\frac{1}{2}$ | B) $\frac{1}{3}$ |
| C) $\frac{1}{6}$ | D) $\frac{5}{6}$ |

Q9. Two dice are thrown together. What is the probability of getting a doublet (same number on both)?

- | | |
|-------------------|-------------------|
| A) $\frac{1}{36}$ | B) $\frac{1}{12}$ |
| C) $\frac{1}{6}$ | D) $\frac{1}{4}$ |

Q10. Two dice are thrown. What is the probability that the sum of the numbers is 10?

- | | |
|-------------------|------------------|
| A) $\frac{1}{12}$ | B) $\frac{1}{6}$ |
| C) $\frac{1}{4}$ | D) $\frac{1}{9}$ |

Q11. From a deck of 52 cards, a card is drawn. What is the probability it is an Ace?

- | | |
|-------------------|-------------------|
| A) $\frac{1}{13}$ | B) $\frac{1}{52}$ |
| C) $\frac{4}{13}$ | D) $\frac{1}{26}$ |

Q12. What is the probability of drawing a red face card from a deck of 52 cards?

- | | |
|-------------------|-------------------|
| A) $\frac{3}{13}$ | B) $\frac{3}{26}$ |
| C) $\frac{1}{26}$ | D) $\frac{1}{13}$ |

Q13. A card is drawn from 52 cards. What is the probability that it is a spade?

- | | |
|--------------------|------------------|
| A) $\frac{1}{13}$ | B) $\frac{1}{4}$ |
| C) $\frac{13}{52}$ | D) Both B and C |

Q14. A bag contains 3 red balls and 5 black balls. A ball is drawn randomly. What is the probability it is red?

- | | |
|------------------|------------------|
| A) $\frac{3}{8}$ | B) $\frac{5}{8}$ |
| C) $\frac{3}{5}$ | D) $\frac{5}{3}$ |

Q15. A box contains 5 red, 8 white, and 4 green marbles. One marble is drawn. Probability it is NOT green?

- | | |
|-------------------|--------------------|
| A) $\frac{4}{17}$ | B) $\frac{13}{17}$ |
| C) $\frac{5}{17}$ | D) $\frac{8}{17}$ |

Q16. What is the probability that a non-leap year has 53 Sundays?

- | | |
|------------------|------------------|
| A) $\frac{1}{7}$ | B) $\frac{2}{7}$ |
| C) $\frac{3}{7}$ | D) $\frac{5}{7}$ |

Q17. A letter is chosen from 'PROBABILITY'. What is the probability it is a consonant?

- | | |
|-------------------|-------------------|
| A) $\frac{4}{11}$ | B) $\frac{7}{11}$ |
| C) $\frac{5}{11}$ | D) $\frac{6}{11}$ |

Q18. A number is chosen from 1 to 10. Probability that it is a multiple of 3?

- | | |
|-------------------|-------------------|
| A) $\frac{3}{10}$ | B) $\frac{1}{5}$ |
| C) $\frac{2}{5}$ | D) $\frac{1}{10}$ |

Q19. If $P(A) = 0.6$ and $P(B) = 0.3$, and A and B are mutually exclusive, what is $P(A \text{ or } B)$?

- | | |
|---------|--------|
| A) 0.18 | B) 0.9 |
| C) 0.3 | D) 0.5 |

Q20. In a family of 2 children, find the probability of having at least 1 girl.

- | | |
|----------|----------|
| A) $1/4$ | B) $1/2$ |
| C) $3/4$ | D) 1 |

Q21. Tickets numbered 1 to 20 are mixed up and one ticket is drawn. Probability it is a multiple of 3 or 5?

- | | |
|-----------|-----------|
| A) $1/2$ | B) $2/5$ |
| C) $8/15$ | D) $9/20$ |

Q22. A box contains cards numbered 6 to 50. A card is drawn. Probability that the number is a perfect square?

- | | |
|-----------|-----------|
| A) $1/45$ | B) $4/45$ |
| C) $1/9$ | D) $5/45$ |

Q23. The sum of probabilities of all elementary events of an experiment is:

- | | |
|------|--------|
| A) 0 | B) 0.5 |
| C) 1 | D) 2 |

Q24. An urn contains 10 black and 5 white balls. Two balls are drawn (with replacement). Probability both are white?

- A) $\frac{1}{9}$
B) $\frac{1}{3}$
C) $\frac{1}{2}$
D) $\frac{2}{9}$

Q25. Probability of a boy being born is $\frac{1}{2}$. What is the probability that a family with 3 children has all boys?

- A) $\frac{1}{8}$
B) $\frac{3}{8}$
C) $\frac{1}{4}$
D) $\frac{1}{2}$

Answer Key

Q1: B	Q2: B	Q3: B	Q4: A	Q5: B
Q6: B	Q7: B	Q8: A	Q9: C	Q10: A
Q11: A	Q12: B	Q13: D	Q14: A	Q15: B
Q16: A	Q17: A	Q18: A	Q19: B	Q20: C
Q21: D	Q22: B	Q23: C	Q24: A	Q25: A